

4.5 Perimeter on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.		 Learning Targets	- I can find the distances between ordered pairs. - I can find the dimensions of a rectangle on a coordinate plane. - I can find the perimeter of a rectangle on a coordinate plane.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas		Working Toward MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane, involving perimeter or area of polygons.	
 Essential Questions	- How can you use what you know about distance on a coordinate plane to find the dimensions of a rectangle? - How can you use what you know about distance on a coordinate plane to find the perimeter of a rectangle on the coordinate plane?			
 Lesson Narrative	Students apply their knowledge of finding distance on the coordinate plane to help them find the dimensions of rectangles, as well as determining the perimeter of a rectangle. Students should be led to see how using the coordinate plane and what we know about the properties of a rectangle give them more than one way to determine distance and dimensions.			
 Component Summary	4.5.1 Warm-Up (~5 min.)	Complete self-assessment on understanding of topics applied in this lesson (<i>SE p. 159</i>)	4.5.4 Collaboration (15 min.)	Collaborate on an error analysis and work together to fix the mistakes (<i>SE p. 162</i>)
	4.5.2 Guided Instruction (10-15 min.)	Find distance between points and identify dimensions of a rectangle on the coordinate plane (<i>SE p. 160</i>)	4.5.5 Your Turn! (5-10 min.)	Find dimensions and solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 164</i>)
	4.5.3 Try It (5-10 min.)	Solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 162</i>)	4.5.6 Wrap-Up (~5-10 min.)	Find dimensions and solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 165</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Perimeter - Absolute value - Coordinate plane - Origin		(Optional) Graph paper	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may always want to add or subtract the coordinates to find the distance without thinking about the location on the coordinate plane, the relation to the axes, and the relation of the points to each other. - Students may believe the orientation of a rectangle is always the same. For example, students may believe length is always the “sides” of the rectangle. - Students may struggle with the concept of absolute value.	Length is not always the longest side of the rectangle. Unless the length is specified or named, students can choose either side as length and width and still be mathematically correct. 4.5.2 Guided Instruction, 4.5.4 Collaboration, 4.5.3 Try It, and 4.5.6 Wrap-Up all specify which side is the length, but 4.5.5. Your Turn! leaves the length open-ended for students to decide. Consider prompting this discussion during 4.5.2 Guided Instruction to prepare students for the flexibility in their 4.5.5 Your Turn! and 4.5.6 Wrap-Up questions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 4.5.4 Collaboration positions students to engage in a Think-Pair-Share. Have students think independently first about Vera and Conor’s work and some errors they might notice before having students discuss with their partner.	MA.K12.MTR.4.1: Discussion and Reflection Students engage in an error analysis where they have to determine which student’s work is correct, if any, and also correct that work using mathematical thinking.
 Differentiate Instruction	Consider moments in which some of the coordinates provided (i.e., a mixed number) could be substituted with a friendlier number to allow students to focus on the process of finding distance and perimeter, rather than the ability to add and subtract with fractions. This will provide additional entry points for students to engage with the content. During 4.5.3 Try It, consider providing students with graph paper as an option to draw and label the fence as an additional visual entry point. During 4.5.4 Collaboration, consider allowing students to explain verbally instead of writing.	
Lesson Notes:		

4.5.1 Warm-Up: Self-Assessment

Component Overview: Students use self-assessment to show understanding of previously learned topics that will be applied in this lesson.



4.5.1 Warm-Up: Self-Assessment

1. Complete each statement to describe your understanding.

- ☐ I know
☐ I need support with
- how to plot rational number

ordered pairs in all four quadrants.

Answers will vary.

- ☐ I know
☐ I need support with
- how to find the perimeter and area

of rectangles.

Answers will vary.

- ☐ I know
☐ I need support with
- how to find the vertical or

horizontal distance between two ordered pairs that are

located in the same or different quadrants.

Answers will vary.



Consider positioning students as experts and create partners based on strengths and gaps of students. For example, identify four students who chose “I know” regarding perimeter and area and three students who chose “I know” for finding distance, and create seven small groups with one of those seven students as the captain of each group. All other students can then opt-in to one of the small groups based on their needs. Alternatively, students can pair up with someone around the room and hold their own small groups using Lesson 4 resources.

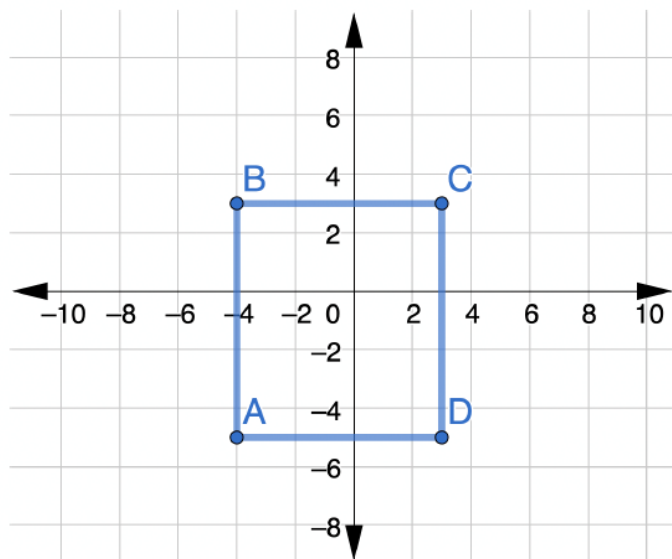
4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles

Component Overview: Students apply what they know about finding distance between points to find the dimensions of a rectangle on the coordinate plane. Using those dimensions, students apply what they know about perimeter of a rectangle to find perimeter on the coordinate plane.



4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles

1. A rectangle has vertices at the points $A(-4, -5)$, $B(-4, 3)$, $C(3, 3)$, and $D(3, -5)$ as shown.



- a. The distance between points A and D is the length of the rectangle. The length of the rectangle is 7 units.

- b. What other two points on the rectangle are the same distance apart as points A and D ?
Points B and C are also the same distance apart as points A and D . They are both 7 units apart.

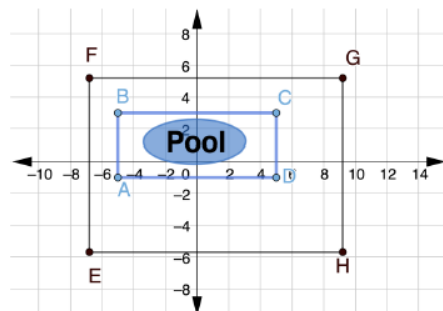


Position students to verbalize how they can find the length and width of the rectangle. Encourage students to recognize that counting the spaces may not necessarily be helpful in this instance because some of the coordinates are not integers. Ensure students understand there is more than one way to find the distance between points on the coordinate plane.

This lesson is a foundation for students understanding how to find slope as the change in the vertical distance over the change in the horizontal distance.

4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles (continued)

- c. Between which two points is the line segment that represents the width of the rectangle?
Answers will vary. The distance between points A and B represent the width. The distance between points D and C represent the width.
- d. Find the width of the rectangle.
8 units
2. Sonja's family installed a new pool. Sonja's dad wants to put up a fence, shown as a blue rectangle, around the pool for safety. The scale of the grid is represented in meters (m). He needs to know how much fencing he will need.



The rectangle that represents the fence has vertices at $A(-5, -1)$, $B(-5, 3)$, $C(5, 3)$, and $D(5, -1)$.

- a. What is the length of the line segment that connects points A and D?
10 meters
- b. What is the length of the line segment that connects points A and B?
4 meters
- c. Explain how to calculate the perimeter of the fence.
Answers will vary. One way to find the perimeter is two times the length from A to D (width of the pool) plus two times the length from A to B (length of the pool).
- d. What is the perimeter of the fence in meters?
28 meters



Always, Sometimes, or Never

Is the length of a rectangle always, sometimes, or never the longest side?

Length is not always the longest side. Unless the question specifies which side is the length, as seen in this question, students can choose either side as length or width and mathematically be correct, as the perimeter does not change.

4.5.5. Your Turn! questions do not specify which side is the length.



If students are struggling to find the length and width of the rectangles, ask probing questions like:

- How do we find distance between two points?
- Are the points on the same side of the axis or different sides? Does this matter? Why or why not?
- *Are the points on the same side of the axis or on opposite sides? How far are the points from the axis? How can we use this to help us find out how far the points are from each other?*



Expressive Language: Eliminate Barriers

Consider providing sentence frames for students to explain their reasoning (i.e., I am able to find the perimeter of the rectangle by ____.)

4.5.3 Try It: Perimeter on the Coordinate Plane

Component Overview: Students solve for perimeter of a rectangle on the coordinate plane.



4.5.3 Try It: Perimeter on the Coordinate Plane

1. Sonja's father plans to put another fence around the yard. The black rectangle on the coordinate grid in the previous component represents the second fence. The scale of the grid is represented in meters (m). The black rectangle has vertices at $E(-6.8, -5.7)$, $F(-6.8, 5.2)$, $G(9.2, 5.2)$, and $H(9.2, -5.7)$.
 - a. What is the length of the line segment that connects point E to point H ?
16 meters
 $|-6.8| + |9.2| = 16 \text{ meters}$
 - b. What is the length of line segment that connects point E to point F ?
10.9 meters
 $|-5.7| + |5.2| = 10.9 \text{ meters}$
 - c. Determine the perimeter of the fence around the yard.
53.8 meters
 $2(16) + 2(10.9) = 32 + 21.8 = 53.8 \text{ meters}$



Prompt students to label the coordinates of the second fence on the graph to help determine the length and width. Consider having some students draw the fence on a piece of graph paper. The visual representation may be helpful as students reason about how to find the perimeter.

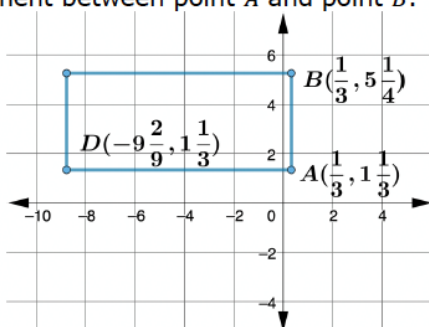
4.5.4 Collaboration: Fix the Mistake

Component Overview: Students engage in a *Fix the Mistake* activity where they look at two different students' work and answers for finding distance and perimeter on the coordinate plane. Students work collaboratively to determine the mistakes and correct them.



4.5.4 Collaboration: Fix the Mistake

- Vera and Conor are working on their homework together to find the perimeter of the rectangle, shown on the coordinate plane. The width of the rectangle is the segment between point A and point B .



Conor counts the units between the numbers, while Vera determines the dimensions by subtracting. Each student's work and answers are shown.

Vera's Thinking	Conor's Thinking
Vera finds the dimension of the width by subtracting the y-coordinates. $5\frac{1}{4} - 1\frac{1}{3} =$ $\frac{21}{4} - \frac{4}{3} =$ $\frac{63}{12} - \frac{16}{12} =$ $\frac{47}{12}$ Vera says the width of the rectangle is $3\frac{11}{12}$ units.	Conor notices there is 1 full unit between point A and point B and two pieces that look like they would combine to make another whole unit. Conor says the width of the rectangle is 2 units.

Discuss each student's work with your partner. Explain any errors that either student made.

Answers will vary. Vera's thinking is correct. Conor forgot that the graph is scaled by a factor of 2 when counting the units.



Think-Pair-Share

Position students to engage in a Think-Pair-Share to discuss each student's work. Students analyze the work independently first and then share their thoughts with a partner before explaining their response in writing. Circulate as students share with a partner and choose one to two students to share with the class the errors they found.

4.5.4 Collaboration: Fix the Mistake (continued)

2. The length of the rectangle is the segment between point A and point D .

Vera finds the dimensions of the length by subtracting the x -coordinates.

$$9\frac{2}{9} - \frac{1}{3} =$$

$$\frac{83}{9} - \frac{3}{9} =$$

$$\frac{80}{9}$$

Vera finds the length of the rectangle to be $8\frac{8}{9}$ units.

Conor counts the units between the numbers again. He counts 4 units between A and D , but then realizes each space on the grid represents 2 units, so he doubles this to 8. He then determines $\frac{2}{9}$ of a unit plus $\frac{1}{3}$ of another unit equals $\frac{5}{9}$. He adds this to the 8.

Conor finds the length of the rectangle to be $8\frac{5}{9}$ units.

Discuss each student's work with your partner. Explain any errors that either student made.

Answers will vary. Conor and Vera both made mistakes in their work. Vera should have added the absolute value of the x -coordinates instead of subtracting them since points A and D are on opposite sides of the y -axis.

Conor made a mistake with the scale again. Although he accounted for the scale when counting the whole units, he did not use the scale when counting the fractional units. He should have added $1\frac{5}{9}$ to 8 to get $9\frac{5}{9}$.

3. After collaborating with each other and understanding their mistakes, Vera and Conor each correctly find the perimeter of the rectangle. What perimeter did they find?

$$\text{Length of rectangle} = 9\frac{5}{9}$$

$$\text{Width of rectangle} = 3\frac{11}{12}$$

$$\text{Perimeter} = 2\left(9\frac{5}{9}\right) + 2\left(3\frac{11}{12}\right) = 26\frac{17}{18} \text{ units}$$



If students are struggling to determine the errors in the students' work, then consider the following prompts:

- What is the scale of the x - and y -axes?
- Why is this important in finding the distance between the points?
- How can you find the distance between points on the coordinate plane?
- How does the location of the points impact how you find the distance between them?
- Does it matter which side of the axes the points are located on? Why?



Consider splitting students into groups and having one group focus on Vera's work and the other on Conor's work. Have the groups be responsible for sharing out what they noticed, and then work together to fix the mistakes. Consider allowing students to explain verbally instead of writing.

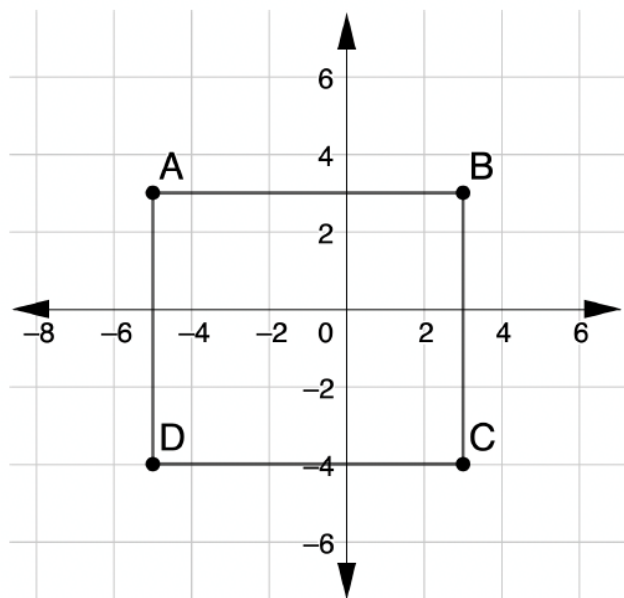
4.5.5 Your Turn!

Component Overview: Students find dimensions of a rectangle on the coordinate plane and use them to find the perimeter of the rectangle.



4.5.5 Your Turn!

- The vertices of a rectangle are at points $A(-5, 3)$, $B(3, 3)$, $C(3, -4)$, and $D(-5, -4)$.



- What is the length, in units, of the rectangle?
8 units
- What is the width, in units, of the rectangle?
7 units
- What is the perimeter, in units, of the rectangle?
 $2(8) + 2(7) = 16 + 14 = 30$ units



Students may still struggle to accurately find the length and width of the rectangle. Encourage students to note the scale of the x - and y -axis. If students still struggle, provide additional scaffolding:

- Are the points on the same side of the axis or on opposite sides?
- How far are the points from the axis?
- How can we use this to help us find out how far the points are from each other?
- Does your answer make sense?
- How can you use the location and relation of the points on the coordinate plane to support your answer?

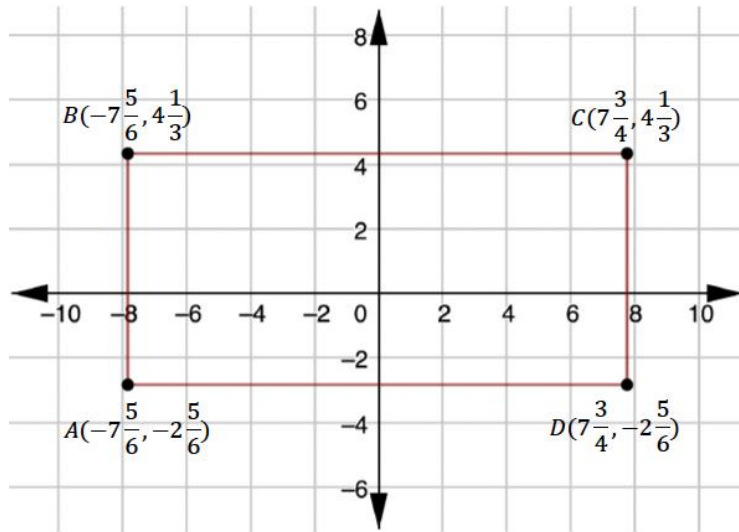


Notice the similarity between this question and the 4.5.6 Wrap-Up, particularly with the open-ended possibility of choosing which side is the length. Consider asking the following questions to the whole group:

- Which side is the length? How do you know?
- Is length always the longest side?
- If the length and width were switched, would the perimeter still be the same?

4.5.5 Your Turn! (continued)

2. Titus wants to put a border around the outside of his garden.



If the scale of the grid is in feet (ft), how many feet of border will he need to buy to completely fence in the garden?

Length of fence (A to B)

$$\left|4\frac{1}{3}\right| + \left|-2\frac{5}{6}\right| = 7\frac{1}{6} \text{ ft}$$

Width of fence (A to D)

$$\left|-7\frac{5}{6}\right| + \left|7\frac{3}{4}\right| = 15\frac{7}{12} \text{ ft}$$

Perimeter of fence

$$2\left(7\frac{1}{6}\right) + 2\left(15\frac{7}{12}\right) = 45\frac{3}{6} \text{ or } 45\frac{1}{2} \text{ ft}$$



Notice that students need to add fractions with different denominators in question 2. Consider substituting these fractions with friendlier numbers, if necessary.

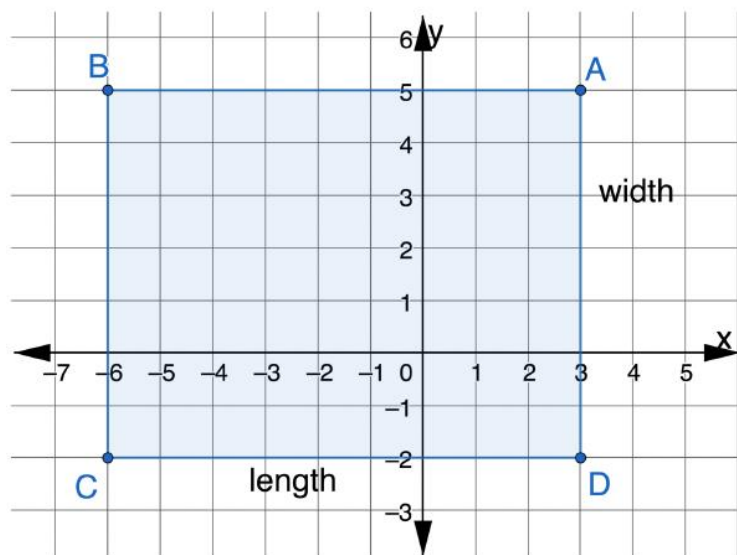
4.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students apply what they have learned to show understanding of finding perimeter of a rectangle on the coordinate plane.



4.5.6 Wrap-Up: Ticket Out the Door

1. A rectangle is shown on the coordinate plane.



Consider asking students to show their work or explain how they found their answer. This information will help inform you as to which students rely on counting and which students use absolute value.

- a. What is the length, in units, of the rectangle?
9 units
- b. What is the width, in units, of the rectangle?
7 units
- c. What is the perimeter, in units, of the rectangle?
 $2(9) + 2(7) = 18 + 14 = 32$ units